RISCV Controller

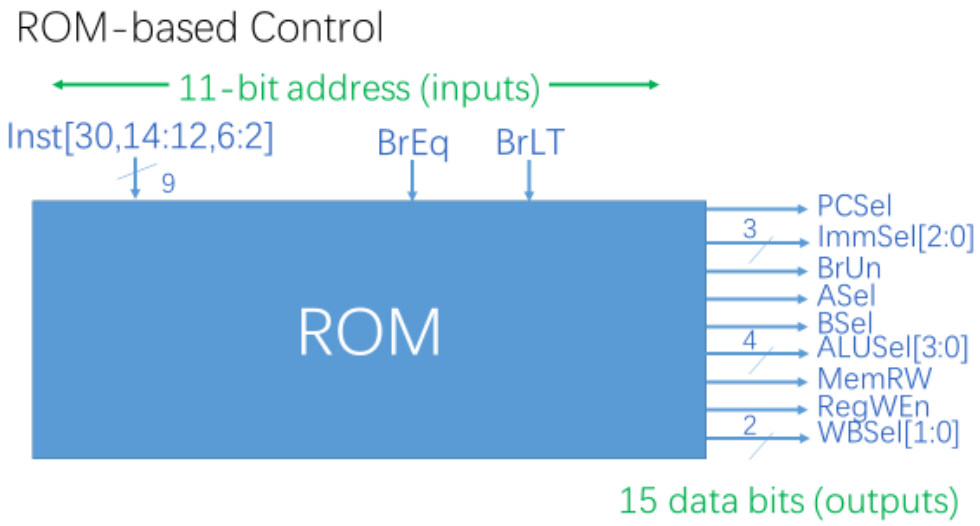


图 3-1 单周期 riscv 核控制器

设计方案

1.Design instructions

This controller is designed with 37 instructions.

imm[31:12]				rd	0110111	U lui
imm[31:12]				rd	0010111	U auipc
imm[20|10:1|11|19:12]				rd	1101111	J jal
imm[11:0]		rs1	000	rd	1100111	I jalr
imm[12|10:5]	rs2	rs1	000	imm[4:1|11]	1100011	B beq
imm[12|10:5]	rs2	rs1	001	imm[4:1|11]	1100011	B bne
imm[12|10:5]	rs2	rs1	100	imm[4:1|11]	1100011	B blt
imm[12|10:5]	rs2	rs1	101	imm[4:1|11]	1100011	B bge
imm[12|10:5]	rs2	rs1	110	imm[4:1|11]	1100011	B bltu
imm[12|10:5]	rs2	rs1	111	imm[4:1|11]	1100011	B bgeu
imm[11:0]		rs1	000	rd	0000011	I lb
imm[11:0]		rs1	001	rd	0000011	I lh
imm[11:0]		rs1	010	rd	0000011	I lw
imm[11:0]		rs1	100	rd	0000011	I lbu
imm[11:0]		rs1	101	rd	0000011	I lhu
imm[11:5]	rs2	rs1	000	imm[4:0]	0100011	S sb
imm[11:5]	rs2	rs1	001	imm[4:0]	0100011	S sh
imm[11:5]	rs2	rs1	010	imm[4:0]	0100011	S sw
imm[11:0]		rs1	000	rd	0010011	I addi
imm[11:0]		rs1	010	rd	0010011	I slti
imm[11:0]		rs1	011	rd	0010011	I sltiu
imm[11:0]		rs1	100	rd	0010011	I xori
imm[11:0]		rs1	110	rd	0010011	I ori
imm[11:0]		rs1	111	rd	0010011	I andi
0000000	shamt	rs1	001	rd	0010011	I slli
0000000	shamt	rs1	101	rd	0010011	I srli
0100000	shamt	rs1	101	rd	0010011	I srai
0000000	rs2	rs1	000	rd	0110011	R add
0100000	rs2	rs1	000	rd	0110011	R sub
0000000	rs2	rs1	001	rd	0110011	R sll
0000000	rs2	rs1	010	rd	0110011	R slt
0000000	rs2	rs1	011	rd	0110011	Rsltu
0000000	rs2	rs1	100	rd	0110011	R xor
0000000	rs2	rs1	101	rd	0110011	R srl
0100000	rs2	rs1	101	rd	0110011	R sra
0000000	rs2	rs1	110	rd	0110011	R or
0000000	rs2	rs1	111	rd	0110011	R and

2.Control Logic Truth Table

RISC-V Control Logic Truth Table											
Inst[]	BrEQ	BrLT	PCSel	Immsel	BrUn	ASel	BSel	ALUsel	MemRW	RegWen	WBSel
R_type 10	*	*	+4(0)	*	*	reg(0)	reg(0)	op	read(0)	1	ALU(2'b01)
add								0000			
sub								0001			
sll								0101			
slt								1001			
sltu								1010			
xor								0010			
srl								0110			
sra								0111			
or								0011			
and								0100			
addi (I_type 这种 9)	*	*	+4(0)	I(3'b000)	*	reg	Imm	op	read(0)	1	ALU(2'b01)
lw （这种 5)	*	*	+4(0)	I(3'b000)	*	reg	Imm	add(4'b0000)	read(0)	1	Mem(2'b00)
lb											
lh											
lbu											
lhu											
S_type 3	*	*	+4(0)	S(3'b001)	*	reg	Imm	add(4'b0000)	write(1)	0	*
sw											
sh											
sb											
B_type 6				B(3'b010)		pc	Imm	add(4'b0000)	read(0)	0	*
beq	1	*	ALU(1)		*						
beq	0	*	+4(0)		*						
bne	1	*	+4(0)		*						
bne	0	*	ALU(1)		*						
blt	*	1	ALU(1)		0						
blt	*	0	+4(0)		0						
bltu	*	1	ALU(1)		1						

bltu	*	0	+4(0)		1						
bge	*	1	+4(0)		0						
bge	*	0	ALU(1)		0						
bgeu	*	1	+4(0)		1						
bgeu	*	0	ALU(1)		1						
jalr	*	*	ALU(1)	I(3'b000)	*	reg	Imm	add(4'b0000)	read(0)	1	PC+4(2'b10)
J_type 1	*	*	ALU(1)	J(3'b011)	*	pc	Imm	add(4'b0000)	read(0)	1	PC+4(2'b10)
jal											
U_type 2	*	*	+4(0)	U(3'b101)	*		Imm		read(0)	1	ALU(2'b01)
lui						*		B(4'b1011)			
auipc						pc		add(4'b0000)			